# Ananya Ramachandra-Krishnamurthy

Computational Bioinformatician & Genomic Data Scientist

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LinkedIn  GitHub  Portfolio

## SUMMARY

Bioinformatics researcher developing deep learning models for genomic variant pathogenicity prediction and multi-omics pipeline orchestration.

## EDUCATION

**Indian Institute of Science, Center for Computational Biology** — B.S. (Research) in Computational Biology in Bioinformatics & Machine Learning  2021 – 2025

## SELECTED WORK

**VariantFlow: Nextflow Whole Genome Sequencing Pipeline**  Lead Developer

High-throughput somatic and germline variant calling pipeline orchestrated on AWS Batch with auto-scaling compute spot instances.

- Processed 300+ patient exome datasets with 99.8% concordance against GIAB gold standards.

Nextflow / GATK / BWA-MEM / Docker

Live Demo · Source Code

## EXPERIENCE

**Summer Genomic Research Fellow, Tata Institute of Fundamental Research (TIFR)**  May 2024 – Jul 2024

Trained transformer models on DNA sequences predicting regulatory motif mutations.

- Outperformed baseline CNN models by 4.2% ROC-AUC on ENCODE transcription factor binding datasets.

## SKILLS

Python / Biopython / R / Bioconductor / PyTorch / Nextflow / Snakemake / Docker / AWS Batch / Scientific Writing / Interdisciplinary Communication